

KENDRIYA VIDYALAYA SANGATHAN, BHOPAL REGION
PRE-BOARD EXAMINATION 2024-25
SET NO. 3

Max. Marks: 80

CLASS-X SCIENCE (086)

Time: 3 hours

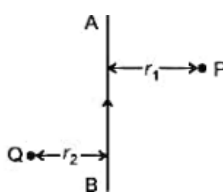
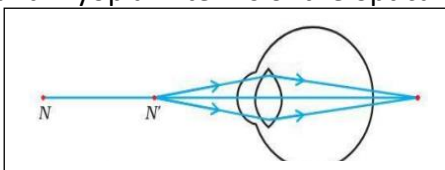
General Instructions:

1. All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2. Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3. Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4. Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5. Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6. Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

SECTION A	
1	Barium chloride on reacting with ammonium sulphate forms barium sulphate and ammonium chloride. Which of the following correctly represents the type of the reaction involved? (i) Displacement reaction (ii) Precipitation reaction (iii) Combination reaction (iv) Double displacement reaction (a) (i) only (b) (ii) only (c) (iv) only (d) (ii) and (iv)
2	A Mendelian experiment consisted of breeding tall pea plants bearing violet flowers with short pea plants bearing white flowers. The progeny all bore violet flowers but almost half of them were short. This suggests that the genetic make-up of the tall parent can be depicted as: (a) TtWW (b) TTWW (c) TTww (d) TtWw
3	An aqueous solution 'A' turns the phenolphthalein solution pink. On addition of an aqueous solution 'B' to 'A', the pink color disappears. Which of the following statement is true for the solution's 'A' and 'B': (a) A is strongly basic and B is a weak base. (b) A is strongly acidic and B is a weak acid. (c) A has a pH greater than 7 and B has a pH less than 7. (d) A has a pH less than 7 and B has a pH greater than 7.
4	Identify 'a', 'b' and 'c' in the following balanced reaction: $\text{"a"} \text{ Pb (NO}_3\text{)}_2\text{(s)} \xrightarrow{\text{Heat}} \text{"b"} \text{ PbO(s)} + \text{"c"} \text{ NO}_2\text{(g)} + \text{O}_2\text{(g)}$ a) 2,2,4 b) 2,4,2 c) 2,4,4 d) 4,2,2
5	Which of the following statements is correct regarding the propagation of light of different colors of white light in air? (a) Red light moves fastest

	(b) Blue light moves faster than green light (c) All the colors of the white light move with the same speed (d) Yellow light moves with the mean speed as that of the red and the violet light	
6	Which of the following is the correct sequence of events of sexual reproduction in a flower? (a) pollination, fertilization, seedling, embryo (b) seedling, embryo, fertilization, pollination (c) pollination, fertilization, embryo, seedling (d) embryo, seedling, pollination, fertilization	[1]
7	Which of the following is an endothermic reaction? (a) Burning of candle. (b) Cooking of food. (c) Decomposition of Vegetable matter. (d) Reaction of Sodium with air	[1]
8	Choose the incorrect statement from the following regarding magnetic lines of field (a) The direction of magnetic field at a point is taken to be the direction in which the north pole of a magnetic compass needle point (b) Magnetic field lines are closed curves (c) If magnetic field lines are parallel and equidistant, they represent zero field strength (d) Relative strength of magnetic field is shown by the degree of closeness of the field lines	[1]
9	Which group of organisms are not constituents of a food chain? (i) Grass, lion, rabbit, wolf (ii) Plankton, man, fish, grasshopper (iii) Wolf, grass, snake, tiger (iv) Frog, snake, eagle, grass, grasshopper (a) (i) and (iii) (b) (iii) and (iv) (c) (ii) and (iii) (d) (i) and (iv)	[1]
10	Ethane - with the molecular formula C_2H_6 has: (a) 9 covalent bonds (b) 8 covalent bonds (c) 7 covalent bonds (d) 6 covalent bonds	[1]
11	Which one of the following can be used as an acid–base indicator by a visually impaired student? (a) Litmus (b) Turmeric (c) Vanilla essence (d) Petunia leaves	[1]
12	In the given food chain, suppose the amount of energy at fourth trophic level is 8kJ, what will be the energy available at the producer level? Grass-----→Grasshopper-----→ Frog-----→ Snake -----→ Hawk (a) 8 k J (b) 80 k J (c) 800 k J (d) 8000 k J	[1]
13	Choose the event that does not occur in photosynthesis: (a) Absorption of light energy by chlorophyll (b) Reduction of carbon dioxide to carbohydrates (c) Oxidation of carbon-to-carbon dioxide (d) Conversion of light energy to chemical energy	[1]
14	Which of the following adaptations in herbivores helps in digestions of cellulose? (a) Longer large intestine (b) Smaller large intestine	[1]

	(c) Smaller small intestine (d) Longer small intestine	
15	Aluminum is used for making cooking utensils. Which of the following properties of aluminum are responsible for the same: (i) Good thermal conductivity (ii) Good electrical conductivity (iii) Ductility (iv) High melting point (a) (i) and (ii) (b) (i) and (iii) (c) (ii) and (iii) (d) (i) and (iv)	[1]
16	_____ conducts impulse from peripheral body parts to brain. a) sensory nerve b) motor nerve c) inter- neuron d) microglia	[1]
Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true		
17	Assertion (A): The maximum concentration of DDT occurs at the very first trophic level of a food chain. Reason (R): DDT is not a biodegradable chemical.	[1]
18	Assertion (A): A concave mirror always forms an image behind it and the image formed is virtual. Reason (R): According to the sign convention, the focal length of a concave mirror is negative.	[1]
19	Assertion (A): On adding dil. HCl to a test tube containing a substance 'X', a colorless gas is produced which gives a pop sound when a burning match stick is brought near it. Reason (R): In this reaction metal 'X' is displaced by Hydrogen.	[1]
20	Assertion (A): The number of chromosomes in a cell and in a germ cell (gametes) is not the same in any species. Reason (R): When two germ cells combine, they restore the normal number of chromosomes in a species.	[1]
SECTION B		
21	Why do we store silver chloride in dark colored bottles? Name the reaction involved along with the balanced chemical equation.	[2]
22	Differentiate between the basic units of human respiratory & excretory system in terms of structure and function respectively? OR Explain the role of the following enzymes in the process of digestion of food in humans: (i) Salivary amylase (ii) Pepsin (iii) Trypsin (iv) Lipase	[2]
23	Give reason for the following statements: a) Placenta is extremely essential for fetal development. b) Sex selective abortion affects a healthy society.	[2]
24	Avantika needs a lens of power -4.5 D for correction of her vision. (a) What kind of defect in vision is she suffering from?	[2]

	(b) What is the focal length of the corrective lens? (c) What is the nature of the corrective lens?	
25	AB is a current carrying conductor in the plane of the paper as shown in figure: (i) Draw the pattern of magnetic field around AB if current flows in direction of B to A. (ii) State the rule to find out the direction of magnetic field around the conductor AB. 	[2]
26	Revanth buried a waste paper underground. After some days, he digged ground again to see the paper and the paper vanished. When he told this to his teacher, his teacher responded that the paper has been decomposed. What are the two types of wastes produced and justify the role of decomposers in the ecosystem?	[2]
SECTION C		
27	A. What causes the bending of shoots in the plants like sunflower? B. What causes the folding of the leaves in 'Touch me not' plant when touched by hand? C. Compare the movement of growth of the pollen tube towards ovule with the bending of shoots of sunflower plant towards sunlight.	[3]
28	Meenakshi heated ethanol with a compound A in presence of a few drops of concentrated H_2SO_4 and observed a sweet-smelling compound B is formed. When B is treated with sodium hydroxide it gives back ethanol and Sodium acetate. a. Identify A and B. b. Write both the chemical reactions involved and name the reactions.	[3]
29	A. In a family of four individuals, the father possessed long ears and the mother possessed short ears. If the parents had pure dominant and recessive traits respectively, then calculate the ratio of genetic makeup of F₂ generation. Show a suitable cross. B. If father had short ears and the mother had long ears, explain what effect it will have on the ratio of genetic makeup in F₂ generation.	[3]
30	A. What is the fundamental difference between hypermetropia and myopia in terms of the optical experience of a person? B. The diagram below shows a special case of an eye defect. (i) What is the defect that is shown in the figure? (ii) State one cause for such a defect? (iii) Explain with reason which lens can be used to correct the defect. 	[3]
31	Comment on the following statements: (i) Metals like Na, K, Ca and Mg are never found in their free state in nature. (ii) Iron articles are galvanized. (iii) Ionic compounds have high melting & boiling point than covalent compounds.	[3]
32	Give reasons for the following: (a) It is dangerous to touch the live wire of the main supply rather than neutral wire. (b) In household circuit, parallel combination of resistances is used. (c) Using fuse in a household electric circuit is important.	[3]

	OR What is a solenoid? Draw a diagram to show field lines of the magnetic field through and around a current carrying solenoid. State the use of magnetic field produced inside a solenoid. List two properties of magnetic lines of force.	
33	A. State Joules law of heating and write its mathematical expression. B. In a household five fans each of 100W are used for 4 hours and an electric press of 500W for 2 hours every day. Calculate the cost of using the fans and electric press for 60 days if the cost of 1 unit of electrical energy is Rs. 6.5.	[3]
SECTION D		
34	(a) In one of the industrial processes used for manufacture of sodium hydroxide, a gas X is formed as by product. The gas X reacts with lime water to give a compound Y which is used as a bleaching agent in chemical industry. Name the process and identify X and Y by giving the chemical equation of the reactions involved. (b) Name the acid present in ant sting and give its chemical formula. Also give the common method to get relief from the discomfort caused by the ant sting. OR (i) Name the simplest saturated hydrocarbon. Draw its electron dot structure. Which type of bonds exist in this compound! (ii) Differentiate between soaps and detergents. (one point) (iii) What is glacial acetic acid ? (iv) Give the IUPAC name of the first member of Alkene.	[5]
35	(A) Draw ray diagrams showing the image formation by a concave lens when an object is placed: (a) at infinity. (b) between infinity and optical centre O of the lens. (B) State Snell's law with respect to refraction of light. Calculate the refractive index of glass if a ray of light enters from air to the glass . (use speed of light = $3 \times 10^8 \text{ m/s}$; speed of light in glass = $2 \times 10^8 \text{ m/s}$) OR (A) A student focused the image of a candle flame on a white screen using a convex lens. He noted down the position of the candle screen and the lens as under: Position of candle = 12.0 cm Position of convex lens = 50.0 cm Position of the screen = 88.0 cm (i) What is the focal length of the convex lens? (ii) Where will the image be formed if he shifts the candle towards the lens at a position of 31.0 cm? (iii) What will be the nature of the image formed if he further shifts the candle towards the lens? (iv) Draw a ray diagram to show the formation of the image in case (iii) as said above.	[5]

36	A banana plant is growing with the help of suckers. These suckers are small plant stem outgrowths which can be separated from the main plant and planted separately and they will grow into a new plant subsequently. (i) Give the name and type of reproduction. (ii) List two advantages the farmer will have on using this type of reproduction. (iii) If a plant produces only male flower. Explain how this plant will be involved in the process of pollination. (iv) Why is the offspring of a plant not absolutely identical to its parent plant? OR (i) Distinguish between pollination and fertilization. (ii) Mention the site and product of fertilization in a flower. (iii) Draw a neat, labelled diagram of a pistil showing pollen tube growth and its entry into the ovule. (any 4 labeling) (iv) what is the fate of ovary and ovule after fertilization.	[5]
SECTION E		
Question No. 37 to 39 are case-based/data -based questions.		
37	Look at Figure 4.1 and answer the following questions: <i>Fig. 4.1</i> (i) What change would you observe in the calcium hydroxide solution taken in tube B? (ii) Write the reaction involved in test tubes A and B respectively. (iii) What will happen if Carbon Dioxide passed in excess? (iv) If ethanol is given instead of ethanoic acid, would you expect the same change? OR (i) Intake of small quantity of methanol can be lethal. Comment (ii) Draw the electron dot structure of ethane(C_2H_6) molecule. (iii) Write the IUPAC name of the following carbon compounds : (a) $CH_3COCH_2CH_2CH_2CH_3$ (c) $CH_3CH_2CH_2CH_2CHO$ (b) $CH_3CH_2CH_2COOH$ (d) CH_3CH_2OH	[4]

38	Read the text given below and answer the questions: Sex determination is the method by which distinction between males and females is established in a species. The sex of an individual is determined by specific chromosomes. These chromosomes are called sex chromosomes or allosomes. X and Y chromosomes are called sex chromosomes. The normal chromosomes other than the sex chromosomes of an individual are known as autosomes. Q1.1 A couple has six daughters. Give the percentage of possibility of their having a girl next (1) time? Q1.2 What is the number of autosomes present in liver cells of a human female is? (1) Q1.3 In XX-XY type of sex determination, which gametes are produced by a human male? (1) Q1.4. Name any two organisms in which sex is not determined genetically. (1) OR Read the text given below and answer the questions: The most obvious outcome of the reproduction process is the generation of individuals of several designs, but in sexual reproduction they may not be exactly alike. The resemblances as well as differences are marked. The rules of heredity determine the process traits and characteristics are reliably inherited. Many experiments have been done to study the rules of inheritance. Q2.1 Which generation will the recessive trait for seed colour and shape of a pea plant will be expressed in a dihybrid cross. (1) Q2.2 Write the phenotype and genotype of all progeny in F₂ generation in a dihybrid cross when a round and yellow seeds are crossed with wrinkled and green seeds. (1) Q2.3 While performing experiments, what is the difference between F₁ and F₂ generation? Explain in context of ratio obtained and characters studied. (2)	[4]
39	Read the following and answer the following questions: In a circuit, several resistors may be combined to form a network. The combination must have two endpoints to connect it with a battery or other elements of the circuit. When the resistors are connected in series then the current flowing in each remains the same but potential differences across each resistor will vary. When the resistances are connected in parallel, the potential difference across each resistor will be the same though a different amount of current will flow in each resistor. Q1.1. Two resistances 10 Ω and 3 Ω are connected in parallel across a battery. If there is a current of 0.2 A in 10 Ω resistor, the voltage supplied by the battery is: (a) 2 V (b) 1 V (c) 4 V (d) 8 V Q1.2. Two wires each having a resistance value equal to R, are first connected in series and then connected in parallel. The plot shows the graphical representation of resistances in both cases. (A) A denotes parallel combination (B) B denotes series combination (C) A denotes series combination and B denotes parallel combination (D) None of the above Q1.3. The household circuits are connected in..... Q1.4. The equivalent resistance of two resistors x and y is "Z", when connected in series and "M" when connected in parallel. Then Z:M is: (a) xy (b) $x + y \times y$ (c) $(x + y)^2/xy$ (d) $xy(2x + 2y)$	[4]

